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THE EGYPTIAN E-LEARNING UNIVERSITY

Fall Semester

2025-2026

Information Technology

IT438

Communication Technology

Kamal Hamza, PhD

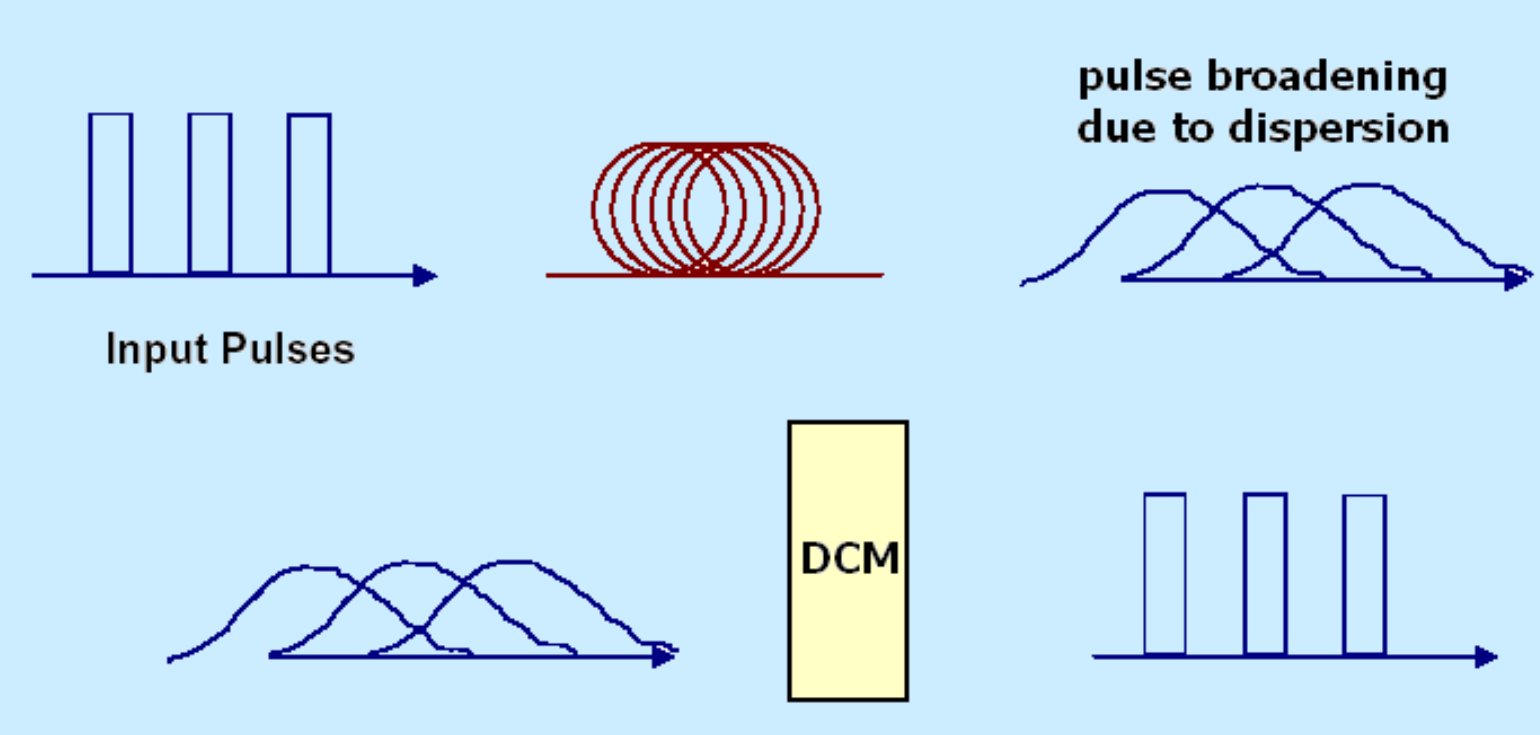
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Acknowledgement: *This presentation contains some figures and text from Data Communications and Networks book by W. Stallings*

Dispersion Compensation Module (DCM)

Dispersion Compensation Module (DCM)

Chromatic dispersion (the spreading of different wavelengths of light) causing signal degradation as it travels through the fiber.



DCM

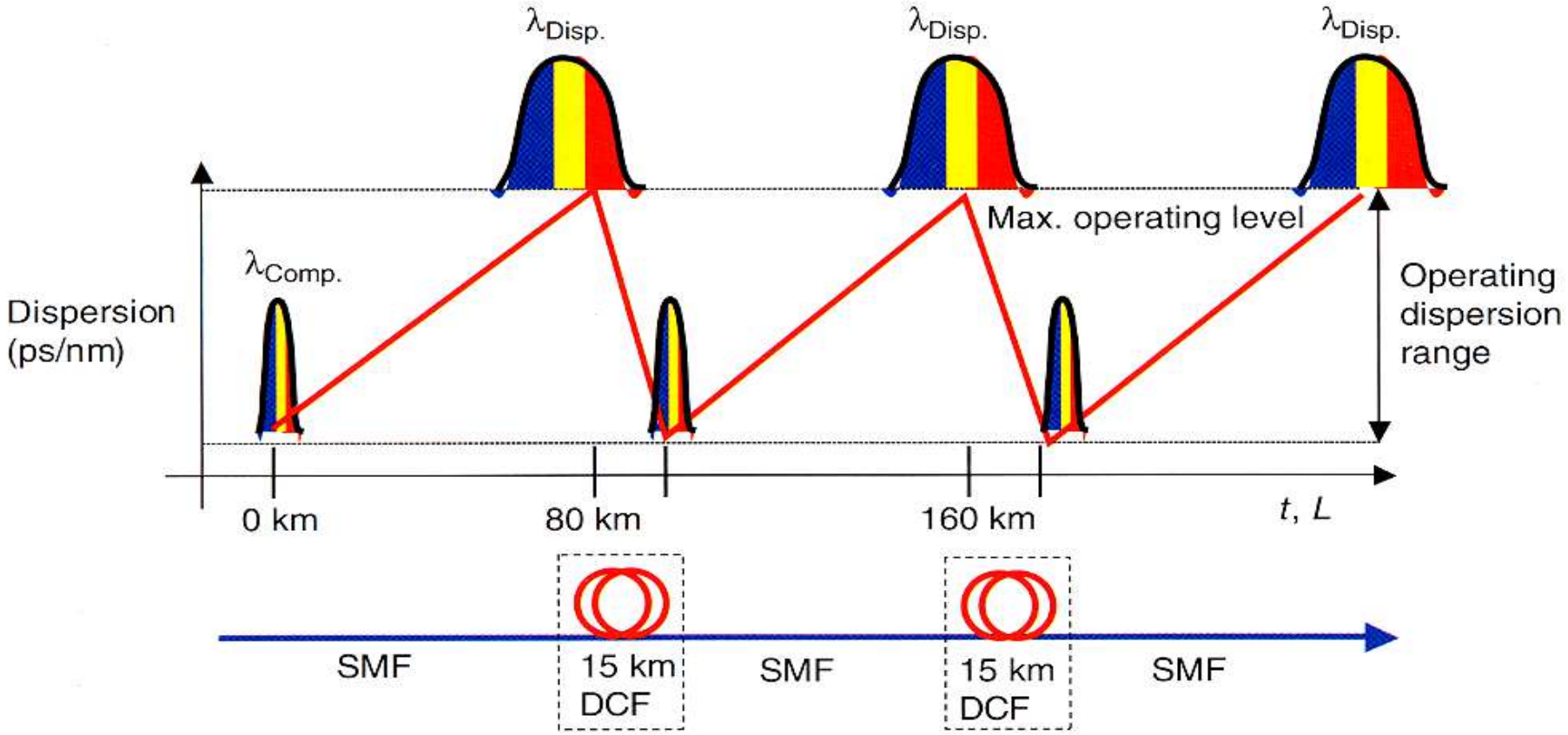
Dispersion Compensating Fiber (DCF)

DCF is engineered with opposite dispersion characteristics, allowing it to reverse this effect when placed in series with regular fibers.

- In SMF, pulses broaden due to positive chromatic dispersion. As different wavelengths spread out, the pulse loses its sharp shape.
- DCF has negative chromatic dispersion, meaning that it compresses the pulse back by reversing the broadening.
- By carefully designing DCF with the right length and dispersion value, it compensates for the dispersion accumulated in the SMF.

DCM

- 80 Km of regular fiber compensating with 15 Km of dispersion compensating fiber.



Optical Amplifiers

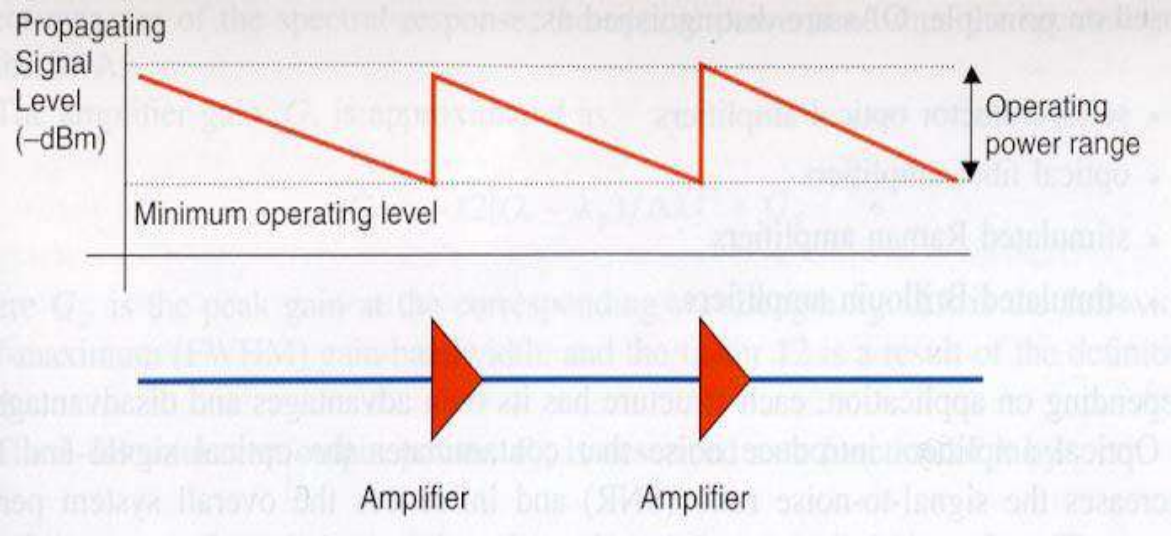
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Optical Amplifier

Types of Optical Amplifier

An optical amplifier is a device that amplifies an optical signal directly without converting it to an electrical signal.

1. Erbium-Doped Fiber Amplifier (EDFA)
2. Raman Amplifier
3. Semiconductor Optical Amplifier (SOA)

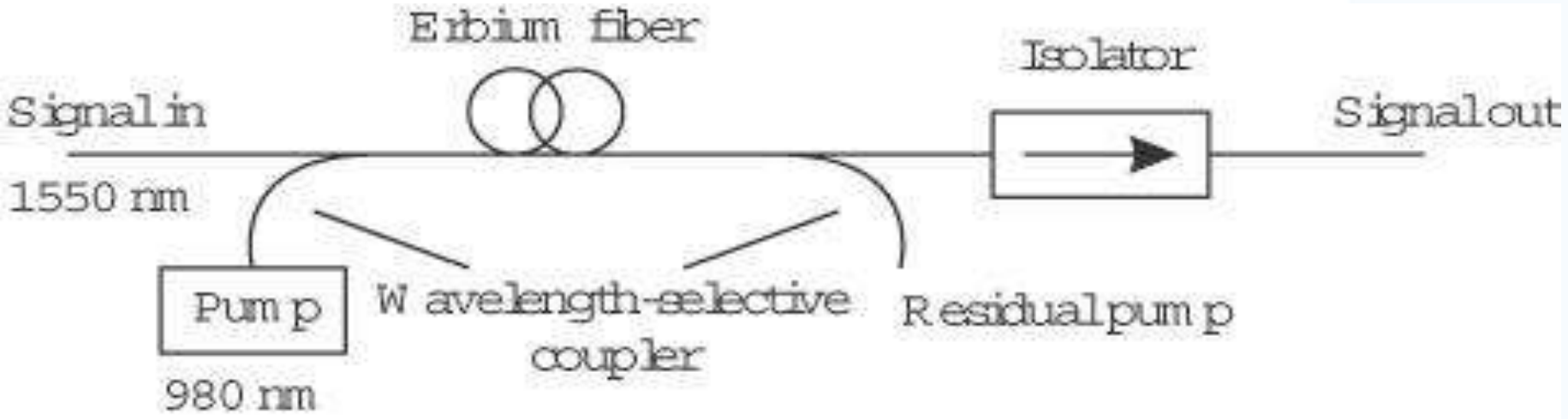


EDFA - Erbium-Doped Fiber Amplifier

EDFA - Erbium-Doped Fiber Amplifier

- EDFAs use a fiber doped with erbium and a pump laser to increase signal strength.
- When a weak signal in the C-band or L-band passes through the erbium-doped fiber, it stimulates the excited erbium ions to release additional photons at the same wavelength, amplifying the original signal.
- High gain with low noise, making them ideal for long-distance communication.
- Compatible with standard single-mode optical fibers.

Erbium-Doped Fiber Amplifier (EDFA)

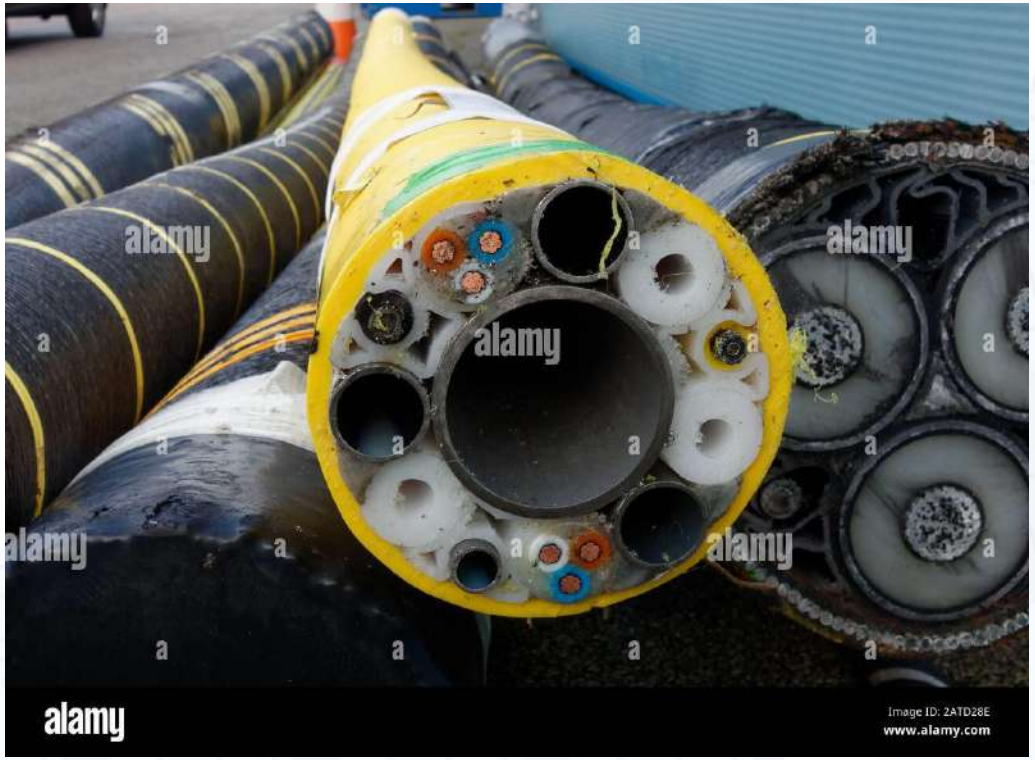


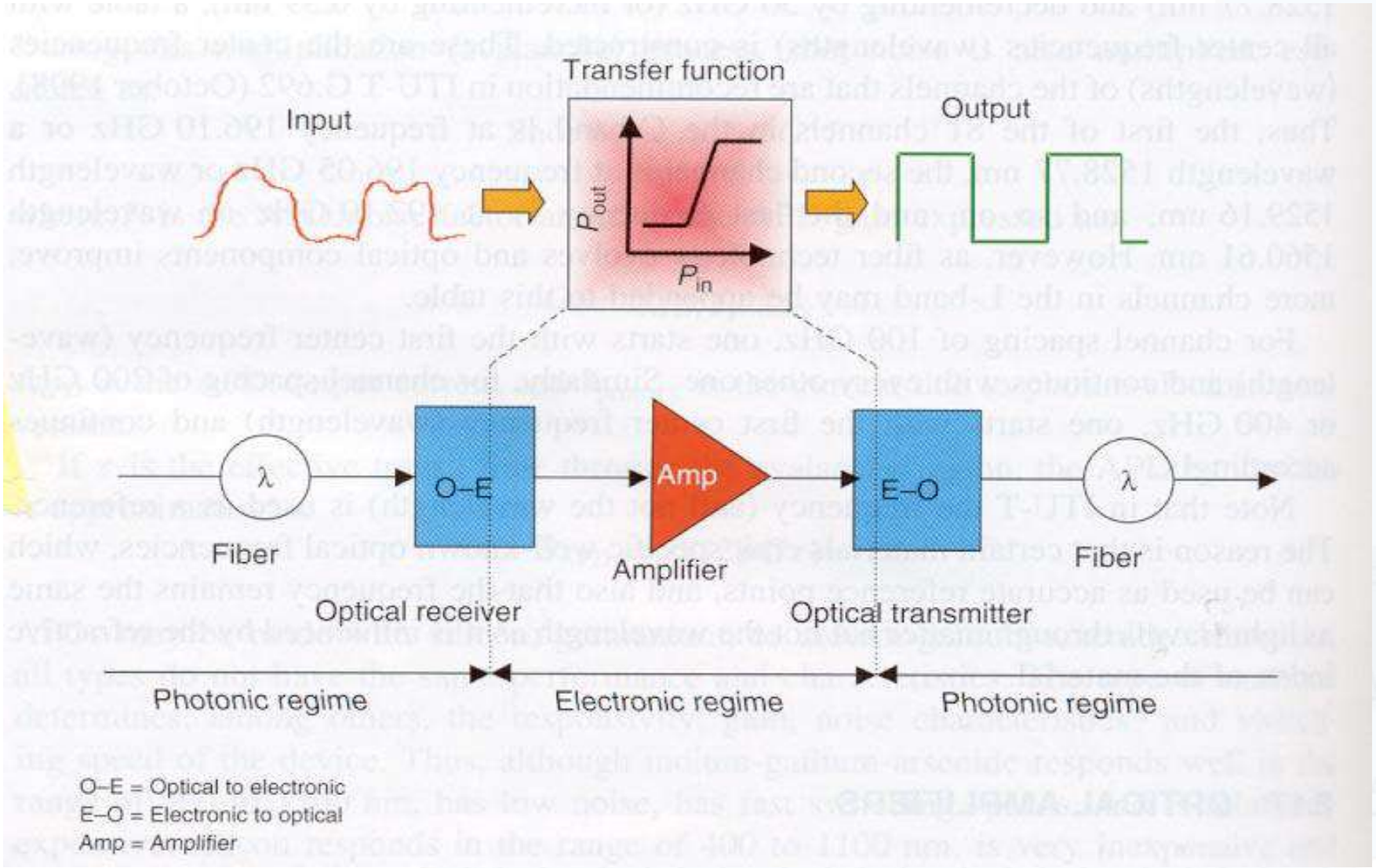
- Length of fiber: core doped with erbium ions Er^{3+}
- Fiber is pumped with a laser at 980 nm or 1480nm.
- Pump is coupled (in- and out-) using a λ-selective coupler
- An isolator is placed at the end to avoid reflections (else this will convert into a laser!)

EDFA - Erbium-Doped Fiber Amplifier (cont.)

Submarine optical cable

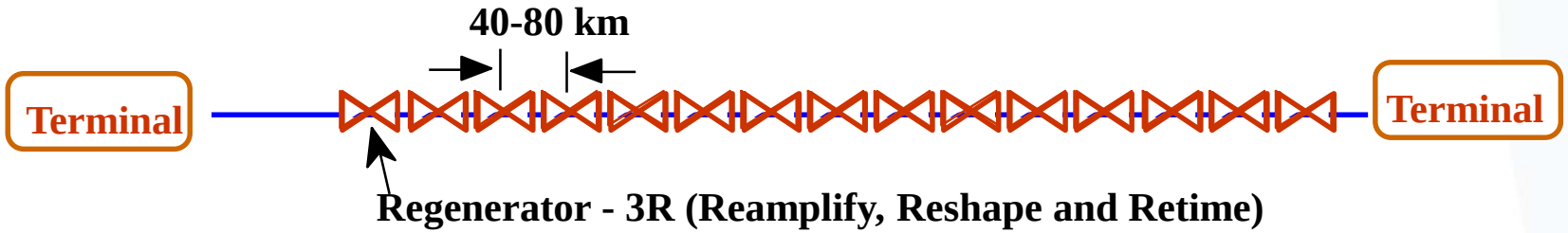


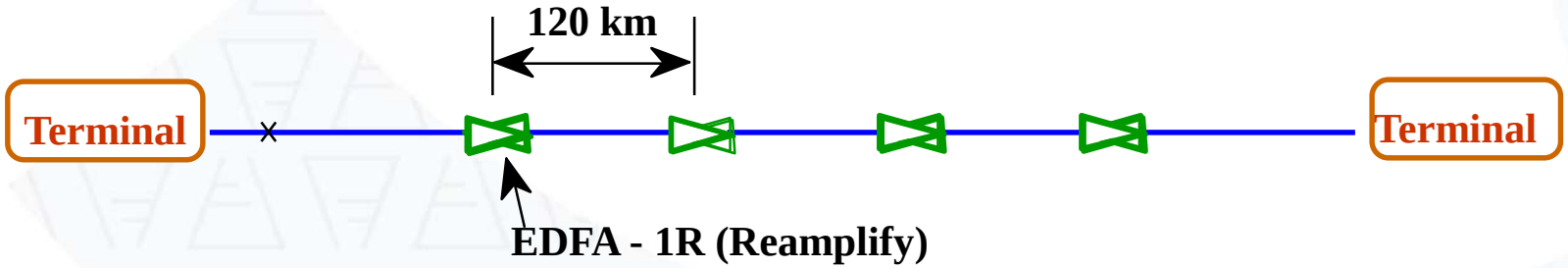


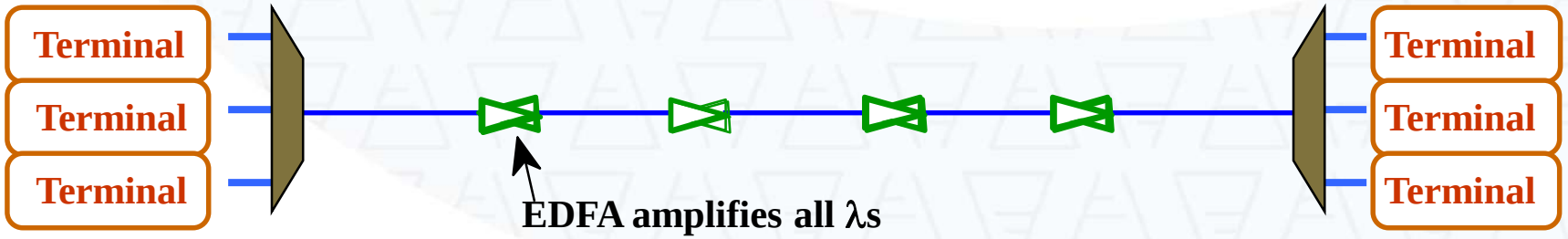


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Optical Amplifiers vs Regenerators







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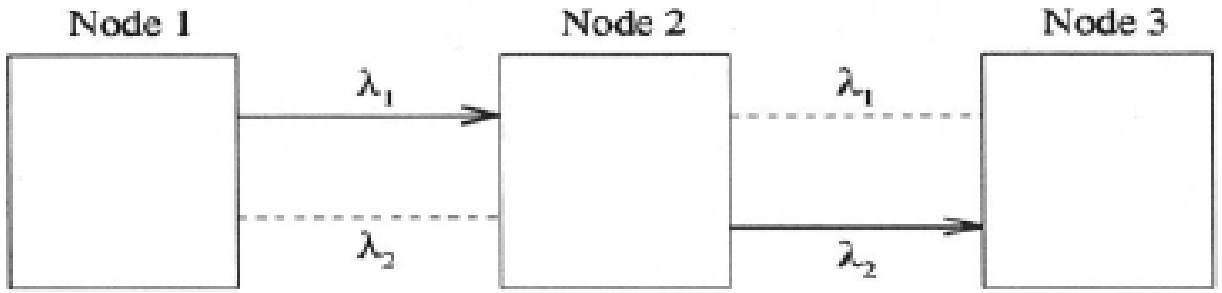
Optical Amplifiers vs Regenerators

Feature	Optical Amplifier	Regenerator (O-E-O)
Mechanism	Amplifies signal directly in optical form	Converts signal to electrical, cleans it, and reconverts to optical
Signal Processing	Amplifies signal and noise together	Removes noise, reshapes, and retimes signal
Signal Quality	Degrades with cumulative noise over distance	Provides a clean, fully restored signal
Application	Long-haul, DWDM systems	Ultra-long-haul, high-integrity systems
Cost and Complexity	Lower, simpler setup	Higher cost and complexity due to O-E-O conversion

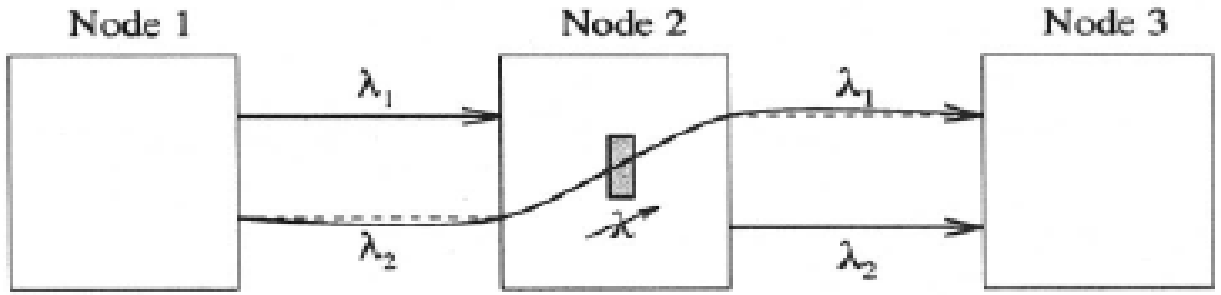
Other Technologies

Wavelength Converters

- Wavelength Converters (WC): converts the wavelength of an input signal to another wavelength.



(a) without converter



(b) with converter

WC Technology

1. **Opto-Electronic Wavelength Conversion (OE-WC)**
   - Incoming optical signal → photodetector converts to electrical signal.
   - Electrical signal is amplified, reshaped, or retimed as needed.
   - Laser tuned to the target wavelength retransmits the processed electrical signal as a new optical signal.
2. **Wavelength Conversion Using Coherent Effects**
   - Relies on nonlinear optical effects, typically in specialized fibers or semiconductor devices.
   - These effects create new wavelengths by combining or modulating existing wavelengths within the fiber, with no electrical conversion involved.

WC Technology

Comparison of OE-WC and Coherent Wavelength Conversion

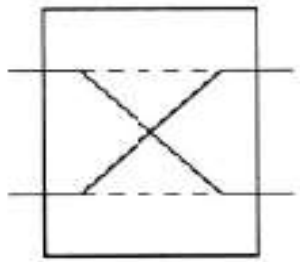
Feature	**Opto-Electronic Wavelength Conversion (OE-WC)**	**Wavelength Conversion Using Coherent Effects**
Conversion Mechanism	Optical to electrical to optical	All-optical, nonlinear interactions
Signal Regeneration	Yes, reshapes and retimes signal	No, only transfers wavelength
Speed	Limited by electronic components	Ultrahigh-speed processing
Applications	Long-haul, metro networks needing regeneration	DWDM, optical packet-switched networks
Signal Quality	High quality due to regeneration	Depends on power management and nonlinear effects
Implementation Complexity	High (requires photodetectors, lasers, electronics)	Moderate to high (requires nonlinear media, careful tuning)
Typical Devices Used	O-E-O transponders, photodetectors, lasers	SOAs, highly nonlinear fiber, Kerr-based devices
Cost	High	Lower to moderat

Optical Switching

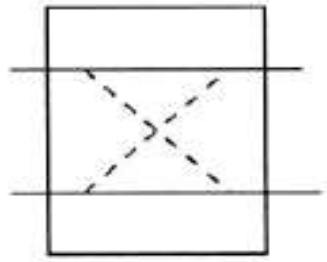
- An optical switch takes an input "optical" signal and direct it to the desired output port.
- As in conventional switching, optical switching can be either: circuit (wavelength-routed) switching, or packet switching (OPS and OBS).
- Currently, WDM optical networks employ circuit switching in the optical domain (instead of routes, we have a route and a wavelength for each connection).
- Packet switching is still face many challenges.

Basic Switching Elements

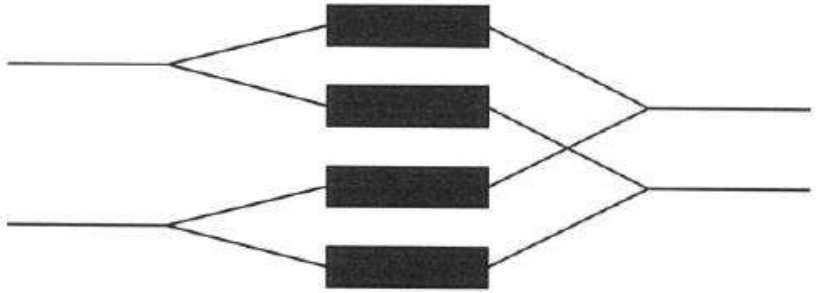
- Extinction Ratio: ratio of output power in ON state to the power in the OFF state
- Insertion loss: fraction of power lost
- Crosstalk: ratio of power at desired vs undesired output



cross state



bar state



Amplifier Gates



THANK YOU

QUESTIONS?





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